4. Animal Classification Using CNN

Objective – To build an image classification model using a pretrained MobileNetV2 network to identify whether an image is a cat, dog, or bird. The aim is to use transfer learning to improve accuracy even with a small dataset.

Libraries / Frameworks used:

TensorFlow (Keras API):
Used to build, train, and evaluate the neural network model.
MobileNetV2:
A pretrained convolutional neural network used for transfer learning to extract image features.
NumPy:
Used for numerical operations and preparing images for prediction.
Matplotlib:
Used to display images and show predicted results.

Pretrained CNN (Transfer Learning) Code –

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

img_size = 224
batch_size = 8

# Load dataset
dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "wildlife_dataset",
    image_size=(img_size, img_size),
    batch_size=batch_size
)

class_names = dataset.class_names
print("Classes:", class_names)

# Prefetch for performance
dataset = dataset.prefetch(buffer_size=tf.data.AUTOTUNE)

# Load pretrained MobileNetV2 (without top layer)
base_model = MobileNetV2(
    input_shape=(224, 224, 3),
    include_top=False,
    weights="imagenet"
)

base_model.trainable = False  # Freeze pretrained layers
```

```
# Build final model
model = models.Sequential([
    layers.Rescaling(1./255),
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(128, activation='relu'),
    layers.Dense(3, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train
model.fit(dataset, epochs=10)

# Evaluate
loss, acc = model.evaluate(dataset)
print("Accuracy:", acc)
```

To Predict:

```
from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

img = Image.open("wildlife_dataset/dog/dog2.jpg").resize((224,224))

img_array = np.array(img)
img_array = np.expand_dims(img_array, axis=0)

prediction = model.predict(img_array)
predicted_class = class_names[np.argmax(prediction)]

plt.imshow(img)
plt.title("Predicted: " + predicted_class)
plt.axis("off")
plt.show()
```

Output –

Program Description:

The program loads images of cats, dogs, and birds and resizes them for training. A pretrained MobileNetV2 model is used to extract important visual features like edges and shapes. The pretrained layers are frozen, and a new output layer is added for classification. The model is then trained using backpropagation to classify the images into three categories.

Result:

The model was successfully trained using transfer learning and achieved better accuracy compared to training from scratch. It was able to correctly classify cat, dog, and bird images, even with a small dataset.